

Sales Performance Analysis

Completed Orders: A total of 40,296 completed orders were recorded after excluding cancelled transactions.

Overall Sales Performance

Metric	Value
Total Revenue	20,608,700.89
Total Quantity Sold	11,348,116 units
Completed Orders	40,296
Average Order Value	511.43
Average Basket Size	281.62 units per order

Highest Revenue Transaction: The highest-value transaction generated **168,469.60** in revenue. This transaction occurred in December 2011.

Customer Analytics: Pareto Analysis

Customers were ranked by total revenue and segmented into the **Top 20%** and **Bottom 80%** using the NTILE() window function.

Segment	Customers	Revenue Contribution	Avg. Customer Revenue	Avg. Orders	Repeat Rate
Top 20%	1,177	77.27%	11,487.78	18.65	99.32%
Bottom 80%	4,704	22.73%	845.33	3.19	65.60%

The results demonstrate a strong Pareto distribution within the customer base. Although only 20% of customers belonged to the highest-value segment, they generated 77.27% of total revenue. These customers also purchased more frequently and exhibited an exceptionally high repeat purchase rate, highlighting their importance to the business.

Behaviour of High-Value Customers

To better understand this high-value customer segment, additional analyses were performed on the top 20% of customers.

Geographic Distribution

Most of the revenue generated by high-value customers originated from the United Kingdom, followed by EIRE, the Netherlands, Germany, and France.

Country	Revenue
United Kingdom	10,143,995.66
EIRE	581,211.59
Netherlands	531,738.82
Germany	341,384.38
France	269,239.61

Product Preferences

High-value customers generated the greatest revenue from the following product categories:

Category	Revenue
Kitchen & Dining	2,967,286.05
Home & Outdoor	2,396,155.63
Textiles & Soft Furnishings	1,646,718.63
Miscellaneous / Unclassified	1,206,839.34
Lighting & Candles	923,463.59

These findings suggest that home-related product categories formed the core of purchasing behaviour among the retailer's highest-value customers.

Seasonal Purchasing Patterns

Revenue generated by high-value customers followed a similar seasonal pattern to the overall business.

The strongest months were:

Month	Revenue
November	1,635,706.33
October	1,395,510.40
September	1,302,584.65
December	1,248,566.81

November recorded the highest cumulative revenue, while February generated the lowest. This pattern reinforces the importance of the final quarter of the year for high-value customer spending.

Weekly Purchasing Behaviour

Purchasing activity among high-value customers was concentrated on weekdays.

Day	Revenue
Thursday	2,812,630.40
Tuesday	2,425,690.83
Wednesday	2,254,887.42
Monday	2,058,585.72

Saturday recorded minimal revenue, indicating that purchasing activity occurred almost entirely during the standard working week.

Time-of-Day Purchasing Behaviour

High-value customers primarily placed orders during normal business hours.

Hour	Revenue
12:00	1,986,779.57
10:00	1,732,341.21
13:00	1,685,084.52
11:00	1,669,712.41

Revenue declined steadily after **16:00**, with very little activity during the evening or early morning hours.

Key Findings

The customer analysis reveals that business performance is heavily influenced by a relatively small group of loyal, high-value customers. These customers contribute over three-quarters of total revenue, purchase substantially more frequently than the rest of the customer base, and demonstrate an exceptionally high repeat purchase rate. Their purchasing behaviour is concentrated within the United Kingdom, focused on home-related product categories, and follows clear seasonal and temporal patterns, with peak activity occurring during the fourth quarter and standard business hours.

Product Performance Analysis

Revenue by Product Category

Product performance was evaluated by analysing revenue generated across product categories and subcategories.

The highest revenue-generating product subcategories are presented below.

Category	Subcategory	Total Revenue
Kitchen & Dining	Drinkware	1,232,668.09
Textiles & Soft Furnishings	Bags (Fabric/Fashion)	1,144,677.73
Home & Outdoor	Ornamental Objects	1,033,572.24

The Kitchen & Dining category featured prominently among the highest revenue-generating products, with both Drinkware and Baking & Cake Supplies exceeding one million in revenue. Strong revenue performance was also observed for Home & Outdoor, Textiles & Soft Furnishings, and Seasonal & Holiday products.

Products with the Highest Sales Volume

Product popularity was also assessed using total quantity sold.

Category	Subcategory	Quantity Sold
Kitchen & Dining	Drinkware	671,036
Home & Outdoor	Ornamental Objects	617,921
Textiles & Soft Furnishings	Bags (Fabric/Fashion)	613,322
Seasonal & Holiday	Christmas	594,921

Several of the highest-volume products also appeared among the highest revenue-generating categories, indicating strong demand across multiple product groups. Drinkware, Ornamental Objects, Bags (Fabric/Fashion) and Christmas products consistently ranked highly for both revenue and quantity sold.

Most Frequently Cancelled Products

Cancellation frequency was analysed to identify the products generating the largest number of cancelled transaction lines.

Category	Subcategory	Cancelled Transactions
Kitchen & Dining	Baking & Cake Supplies	1,582
Kitchen & Dining	Drinkware	1,515
Non-Merchandise	Transaction/Fee/Adjustment	1,084
Lighting & Candles	Candles & Candle Holders	934

Products generating the largest number of cancellations largely mirrored the retailer's highest-selling product groups. This suggests that higher cancellation counts are partly a consequence of greater sales volume rather than poor product performance alone.

Product Cancellation Rate

To provide a fairer assessment of cancellation behaviour, cancellation rates were calculated by dividing cancelled transaction lines by total transaction lines for each product subcategory.

Category	Subcategory	Cancellation Rate
Lighting & Candles	Decorative Lights	6.79%
Kitchen & Dining	Children's Tableware	6.50%
Kitchen & Dining	Popcorn & Snack Accessories	4.79%
Home & Outdoor	Vases & Planters	4.40%

Unlike cancellation frequency, cancellation rate identifies products that are proportionally more likely to be cancelled regardless of sales volume.

Among physical merchandise, Decorative Lights, Children's Tableware, and Popcorn & Snack Accessories recorded the highest cancellation rates and may warrant further investigation into inventory management, order fulfilment processes, or customer satisfaction.

Cancellation analysis also shows that while popular products naturally recorded higher numbers of cancelled transactions because of their sales volume, the cancellation rate identified specific product groups that were disproportionately affected by cancellations.

Market and Sales Trends Analysis

Revenue by Country

Geographical analysis was conducted to identify the retailer's highest revenue-generating markets.

Country	Total Revenue
United Kingdom	17,524,246.80
EIRE	665,700.47
Netherlands	557,541.60
Germany	426,126.46
France	352,130.73

The United Kingdom dominated revenue generation, contributing substantially more revenue than every other country. Among international markets, EIRE, the Netherlands, Germany, and France represented the retailer's strongest sources of revenue.

These findings indicate that the retailer's sales were primarily concentrated within its domestic market while maintaining a smaller but meaningful customer base across several European countries.

Customer Distribution by Country

Customer distribution was analysed to determine the retailer's geographic reach.

Country	Customers
United Kingdom	5,353
Germany	107
France	95
Spain	41
Belgium	29

The United Kingdom accounted for most customers, reinforcing its importance as the retailer's primary market. Germany and France represented the largest international customer bases, although they remained considerably smaller than the domestic customer population.

When compared with the revenue analysis, the results suggest that countries with larger customer populations generally generated higher total revenue.

Monthly Revenue Trends

Monthly revenue was analysed across the three years covered by the dataset to identify seasonal sales patterns.

Revenue consistently increased during the final quarter of the year, particularly in October and November.

Across both complete years in the dataset:

- November generated the highest monthly revenue.
- October consistently ranked among the strongest-performing months.
- Revenue generally declined during January and February before recovering throughout the remainder of the year.

This recurring pattern provides strong evidence of seasonal demand, with customer spending peaking during the months preceding the holiday shopping period.

Cancellation Patterns by Weekday

Cancellation behaviour was analysed across the days of the week.

Weekday	Cancellation Rate
Thursday	2.34%
Tuesday	2.16%
Friday	1.95%
Wednesday	1.87%
Monday	1.65%
Sunday	0.82%
Saturday	0.00%

The highest cancellation rates occurred on Thursday and Tuesday, while weekend cancellation rates were considerably lower.

However, these findings should be interpreted alongside transaction volume. Weekdays also recorded the largest number of sales.

Key Findings

The retailer's performance was heavily concentrated within the United Kingdom, which accounted for the overwhelming majority of both customers and revenue. Nevertheless, several European countries—including EIRE, the Netherlands, Germany, and France—made meaningful contributions to international sales.

The analysis also demonstrates clear evidence of seasonality. Revenue and product demand consistently peaked during October and November, indicating a strong pre-holiday sales period. Finally, while weekday cancellations were slightly more common than weekend cancellations although this might be a reflection of the transaction volumes processed during the working week.